Imagination Digitalized using EEG

A Project Report Submitted by

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in partial fulfillment of the requirements for the award of the degree of

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Declaration

I hereby declare that the work presented in this Project Report titled Imagination Digitalized using EEG submitted to the Indian Institute of Technology (IIT), Jodhpur in partial fulfilment of the requirements for the award of the degree of Master of Technology, is a bonafide record of the research work carried out under the supervision of Dr. Vignesh Muralidharan, Professor, School of Artificial Intelligence and Data Science (SAIDE). The contents of this Project Report in full or in parts, have not been submitted to, and will not be submitted by me to, any other Institute or University in India or abroad for the award of any degree or diploma.

Signature

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Certificate

This is to certify that the Project Report titled Imagination Digitalized using EEG, submitted by Manoj Kumar Tiwari (M25AI1071) to the Indian Institute of Technology (IIT), Jodhpur for the award of the degree of Master of Technology, is a bonafide record of the research work done by him under my supervision. To the best of my knowledge, the contents of this report, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

Signature

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Abstract

Decoding mental imagery from electroencephalography (EEG) signals and translating it into photorealistic visual output represents one of the most ambitious frontiers in brain-computer interface research. This work addresses the problem directly: given raw multi-channel EEG recordings collected while subjects imagine visual objects, can a deep learning system reconstruct corresponding images? Two end-to-end pipelines were built and evaluated on a 10-class cross-subject visual imagery benchmark. Pipeline 2 uses a frozen CSBrain EEG encoder, an EEGProjection MLP, and TinyLlama-1.1B with LoRA fine-tuning to generate text descriptions that drive Stable Diffusion 2.1. Pipeline 3 trains a compact EEGCLIPMapper with Q-Former-style cross-attention to directly align EEG features with CLIP embedding space, using a hybrid contrastive-classification objective. Both pipelines substantially exceed the 10% chance baseline; Pipeline 3 achieves 35.8% test accuracy versus 32.6% for Pipeline 2. A key finding is that CLIP image embeddings of stimulus photographs (mean pairwise cosine similarity ≈ 0.54) are far more effective contrastive training targets than text embeddings (≈ 0.90).

Keywords: EEG, Brain-Computer Interface, Visual Imagery Decoding, Stable Diffusion, CLIP, Contrastive Learning

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1 INTRODUCTION

1.1 Background and Motivation

At some fundamental level, the human brain is always generating things. Close your eyes and picture a golden retriever — your visual cortex lights up in ways that actually overlap with what happens when you look at a real dog. Neuroscientists have known about this for a long time, but building a machine that can exploit it is a different story. The difficulty is not conceptual; it is technical. EEG signals are messy, vary wildly between people, and are high-dimensional in ways that make them hard to work with even with modern tools.

EEG has been the go-to modality for brain-computer interface (BCI) work for decades. Not because it gives a detailed view of what is happening in the brain — it really does not — but because it is cheap, non-invasive, and can track neural dynamics at millisecond timescales. The downside is that EEG picks up the combined electrical activity of huge populations of neurons, all blurred by the skull and scalp tissue sitting between the brain and the electrodes. Getting anything useful out of that for classification is hard enough; building an image generation system on top of it is even harder.

That said, the last few years have seen real progress on both fronts. On the EEG side, large foundation models trained with self-supervised objectives have shown they can learn transferable representations from raw recordings. On the generation side, diffusion models like Stable Diffusion produce images that would have seemed impossible just a few years back. This project is trying to connect these two lines of progress: given raw EEG recordings collected while subjects imagine visual objects, can a deep learning system reconstruct corresponding images?

1.2 Problem Statement

Given a multi-channel EEG recording captured while a subject imagines a visual object drawn from a fixed set of categories, the task is to generate a photorealistic image of that imagined object. Formally, we have a set of EEG trials $\{(\mathbf{x}_i, y_i)\}_{i=1}^{N}$ where $\mathbf{x}_i \in \mathbb{R}^{C \times T}$ is a multi-channel time-series recording with $C = 32$ channels and $T = 800$ time samples (at 200 Hz after downsampling), and $y_i \in \{0, 1, \ldots, 9\}$ is a class label indicating which of 10 visual objects was imagined. The goal is to learn a function $f : \mathbb{R}^{C \times T} \to \mathcal{I}$ that maps raw EEG recordings to images $\mathcal{I}$ that are semantically consistent with the imagined object.

This formulation reveals two intertwined sub-problems: (1) *decoding*, which asks which object the

EEG signal corresponds to, and (2) *synthesis*, which asks how to generate a plausible photorealistic image of that object. A key design insight pursued in this project is that keeping these sub-problems at least partially separate — rather than forcing an end-to-end mapping from waveforms to pixels — leads to more tractable and interpretable systems.

1.3 Objectives

The main objectives of this project were:

1. To develop an EEG-to-image pipeline (Pipeline 2) operating through a language model intermediary, using a frozen CSBrain foundation model encoder, an EEG-to-LLM projection MLP, and TinyLlama-1.1B with LoRA fine-tuning to generate text descriptions that drive Stable Diffusion 2.1 image synthesis.

2. To design and train a compact EEGCLIPMapper network (Pipeline 3) that directly aligns EEG features with CLIP embedding space using a hybrid contrastive-classification objective, producing conditioning signals for Stable Diffusion v1.5.

3. To perform a rigorous cross-subject evaluation on the 10-class Figshare Visual Imagery EEG dataset, comparing both pipelines against the 10% chance baseline and against each other.

4. To analyse the role of CLIP image embeddings versus text embeddings as contrastive training targets, and characterise the importance of two-phase training strategy for stable convergence of the CLIP mapper.

1.4 Scope

This project focuses on the Figshare Visual Imagery EEG dataset containing recordings from 22 subjects across 10 visual categories spanning three semantic domains: animals (dog, bird, fish), geometric figures (pentagram, square, circle), and everyday objects (scissors, watch, cup, chair). The evaluation protocol is strictly cross-subject: training on 16 subjects (sub-01 to sub-16), validation on 3 subjects (sub-17 to sub-19), and testing on 3 unseen subjects (sub-20 to sub-22). This reflects the real-world BCI requirement of generalising to individuals not seen during training.

The project does not include real-time BCI inference, user studies, or subject-specific calibration. It is an offline deep learning research study comparing two end-to-end generation pipelines.

1.5 Contributions

The main contributions of this work include:

- A complete and reproducible implementation of two end-to-end EEG-to-image pipelines on the Figshare Visual Imagery EEG dataset, both building on a frozen CSBrain EEG foundation model as the encoder backbone.
- An EEGCLIPMapper architecture that bridges EEG token features to CLIP embedding space using a Q-Former-style cross-attention mechanism, trained with a hybrid composite loss ($\mathcal{L} = 5.0\,\mathcal{L}_{\text{cls}} + 2.0\,\mathcal{L}_{\text{cont}} + 1.0\,\mathcal{L}_{\text{cos}} + 0.1\,\mathcal{L}_{\text{mse}}$) and achieving 35.8% cross-subject test accuracy on a 10-class benchmark.
- An empirical demonstration that CLIP image embeddings of stimulus photographs (mean pairwise cosine similarity ≈ 0.54) are substantially more discriminative contrastive targets than CLIP text embeddings (≈ 0.90) for this visual imagery decoding problem.
- A two-phase training strategy (warmup with classification and contrastive losses only, followed by full joint training with MSE alignment) that stabilises training by preventing conflicting gradient directions during early optimisation.

1.6 Report Organization

This report is organized as follows: Section 2 presents a comprehensive review of related work on EEG processing, foundation models, brain decoding, and generative models. Section 3 describes the methodology, including problem formalisation, the dataset, and the mathematical objectives of both pipelines. Section 4 details the implementation of Pipeline 2 (EEG-to-text-to-image) and Pipeline 3 (EEG-to-CLIP). Section 5 presents quantitative results and qualitative analysis. Section 6 discusses findings, limitations, and unexpected observations. Section 7 concludes the report and outlines future work.

2 LITERATURE REVIEW

2.1 EEG Signal Processing and Feature Extraction

The history of EEG signal processing is, in many respects, the history of feature engineering. For decades, the dominant approach to EEG-based classification relied on hand-crafted features extracted from well-understood frequency bands. The alpha band (8–13 Hz) is associated with relaxed wakefulness; the beta band (13–30 Hz) with active thinking and motor planning; gamma oscillations ($>$30 Hz) with high-level cognitive processes. Common Spatial Patterns (CSP) — a supervised algorithm that learns spatial filters maximising the variance ratio between two conditions — became the de facto standard for motor imagery classification in BCIs, particularly after its success in the BCI Competition IV datasets [1].

Event-related (de)synchronisation (ERD/ERS) is another foundational concept. During motor imagery, a characteristic suppression of mu-rhythm (8–12 Hz) and beta activity occurs contralateral to the imagined limb. For visual imagery, the relevant signatures are less well-characterised: posterior alpha suppression, increases in occipital gamma power, and subtle changes in parieto-occipital connectivity have all been reported [2].

More recent work has moved decisively toward end-to-end learned features. EEGNet [3] demonstrated that a small convolutional network with depthwise separable convolutions could outperform CSP on multiple EEG classification benchmarks while remaining compact enough to run on embedded hardware. ShallowConvNet and DeepConvNet [4] explored deeper convolutional architectures, demonstrating that temporal and spatial filters learned jointly from data capture structure that hand-crafted features miss. Transformers made their way into EEG research as well: EEG-Conformer [5] combined local convolutional encoders with global self-attention, and ATCNet [6] mixed attention mechanisms with temporal convolutions, achieving strong results on BCI Competition IV motor imagery benchmarks.

2.2 Foundation Models for EEG

Probably the most relevant recent development for this project is the emergence of EEG foundation models — large models pre-trained on diverse EEG datasets that can be fine-tuned for specific downstream tasks. CSBrain [7], used as the encoder backbone in both pipelines of this work, was trained using a masked patch prediction objective on a large EEG corpus. Its architecture groups electrodes by anatomical region (frontal, fronto-central, central, parietal, occipital) and employs separate modules for within-region and cross-region dynamics, providing neurophysiologically meaningful representations that transfer well to new tasks.

The decision to freeze CSBrain's weights in both pipelines is a practical one: with only 22 subjects doing visual imagery tasks, there is far too little data to train a large encoder from scratch without severe overfitting. Freezing it and only training downstream components allows the model to leverage everything learned during CSBrain's pre-training while keeping the actual number of learnable parameters small.

Other notable foundation models in this space include BENDR [8], which adapts the wav2vec contrastive self-supervised framework for EEG, and LaBraM [9], which performs masked signal modelling across multiple EEG tasks. The area is moving quickly, and foundation model-based approaches clearly dominate the direction of future BCI research.

2.3 Brain Decoding and Neural Image Reconstruction

The dream of reading out what a person is perceiving or imagining from brain activity has a long history. Early work by Haxby et al. [10] showed using fMRI that face and object categories could be decoded from patterns of activity in ventral visual cortex. Kay et al. [11] demonstrated that fMRI voxel patterns could be used to identify which natural image a subject was viewing from a large database — establishing the principle of brain-based image identification.

The transition from identification to *reconstruction* — actually generating an image from brain activity — required the availability of powerful generative models. Shen et al. [12] trained a deep generative network on fMRI data to reconstruct perceived images. Ozcelik and VanRullen [13] achieved a qualitative leap using Stable Diffusion conditioned on brain features. MindEye [14] pushed further by mapping fMRI representations into CLIP space and using Versatile Diffusion for generation, achieving impressive reconstructions on the Natural Scenes Dataset.

On the EEG side, work is more limited due to EEG's poor spatial resolution. Tirupattur et al. [15] used conditional GANs to reconstruct images from EEG on a 40-class dataset. Kavasidis et al. [16] demonstrated that class-level information does exist in EEG signals via GAN-based generation. More recently, Bai et al. [17] tried CLIP-guided diffusion generation from EEG signals and found that contrastive pre-training in CLIP space substantially improves both decoding accuracy and image quality — the closest prior work to Pipeline 3 in this project.

2.4 Comparative Analysis of Image Generation Paradigms

Table 2.1 compares the image generation paradigms used in brain decoding literature across key criteria relevant to this project.

Table 2.1: Comparison of image generation approaches used in neural decoding literature.

Method	**High Quality**	**Stable Training**	**Conditional Gen.**	**Limitation**
GANs [18]	Partial	No	Yes	Mode collapse, instability
VAEs [19]	No	Yes	Yes	Blurry outputs
Diffusion (pixel) [20]	Yes	Yes	Yes	Computationally heavy
Latent Diffusion [21]	Yes	Yes	Yes	Requires VRAM for VAE

GANs [18] were the dominant approach in early EEG-to-image work. While outputs can be sharp, training instability and mode collapse are constant risks. VAEs [19] are more probabilistically principled but produce blurrier outputs due to pixel-level reconstruction loss. Diffusion models [20] have taken over as the state-of-the-art for image quality. Stable Diffusion [21] brings computation to a manageable level by performing the diffusion process in latent space, and Classifier-Free Guidance [22] allows tuning the adherence to the conditioning signal at inference time. Both pipelines in this project use Stable Diffusion as the image generation backend.

2.5 Contrastive Learning and CLIP

The CLIP model [23] is trained on 400 million image-text pairs, learning to align image and text representations in a shared embedding space using a contrastive objective. For brain decoding, CLIP space is a natural target: if an EEG signal can be mapped close to the CLIP embedding of the correct stimulus image, the decoding problem is essentially solved — find the nearest class prototype and generate an image from there.

A finding of particular relevance to this project is that CLIP image embeddings of the 10 stimulus classes are much more spread out than CLIP text embeddings for the same classes (mean pairwise cosine similarity ≈ 0.54 vs. ≈ 0.90). The text embeddings are crowded together and provide almost no useful gradient signal for contrastive training. The image embeddings are more discriminative, and training the EEG mapper to align with them results in embeddings that are better separated by class.

The Q-Former architecture from BLIP-2 [24] and the Perceiver Resampler from Flamingo [25] provide the cross-attention expansion mechanism that bridges between a variable-length modality-specific token sequence and the fixed-length CLIP embedding format, directly inspiring the EEGCLIPMapper design in Pipeline 3.

2.6 Gap Analysis

Based on the literature review, the following gaps are addressed by this project:

- **Gap 1: Limited use of diffusion models for EEG-to-image generation.** Most EEG-to-image work still uses GANs or VAEs, which simply cannot produce the quality of modern diffusion models. This project systematically applies Stable Diffusion to EEG-driven image generation with both a language-mediated and a direct CLIP-alignment approach.

- **Gap 2: Language model-mediated EEG decoding is unexplored.** Using a fine-tuned LLM as an intermediate decoding step between EEG features and image generation (Pipeline 2) has barely been tried, despite LLMs carrying rich prior knowledge about visual concepts that could compensate for weak EEG signals.

- **Gap 3: Most EEG-to-image benchmarks use very few classes.** Prior work typically uses 2–6 classes. This project evaluates on 10 categories across three semantic domains under a strict cross-subject protocol, which is a harder and more realistic test.

This project addresses all three gaps by implementing, comparing, and systematically analysing two complete pipelines on the 10-class cross-subject benchmark, with a focus on understanding what makes each approach succeed or fail.

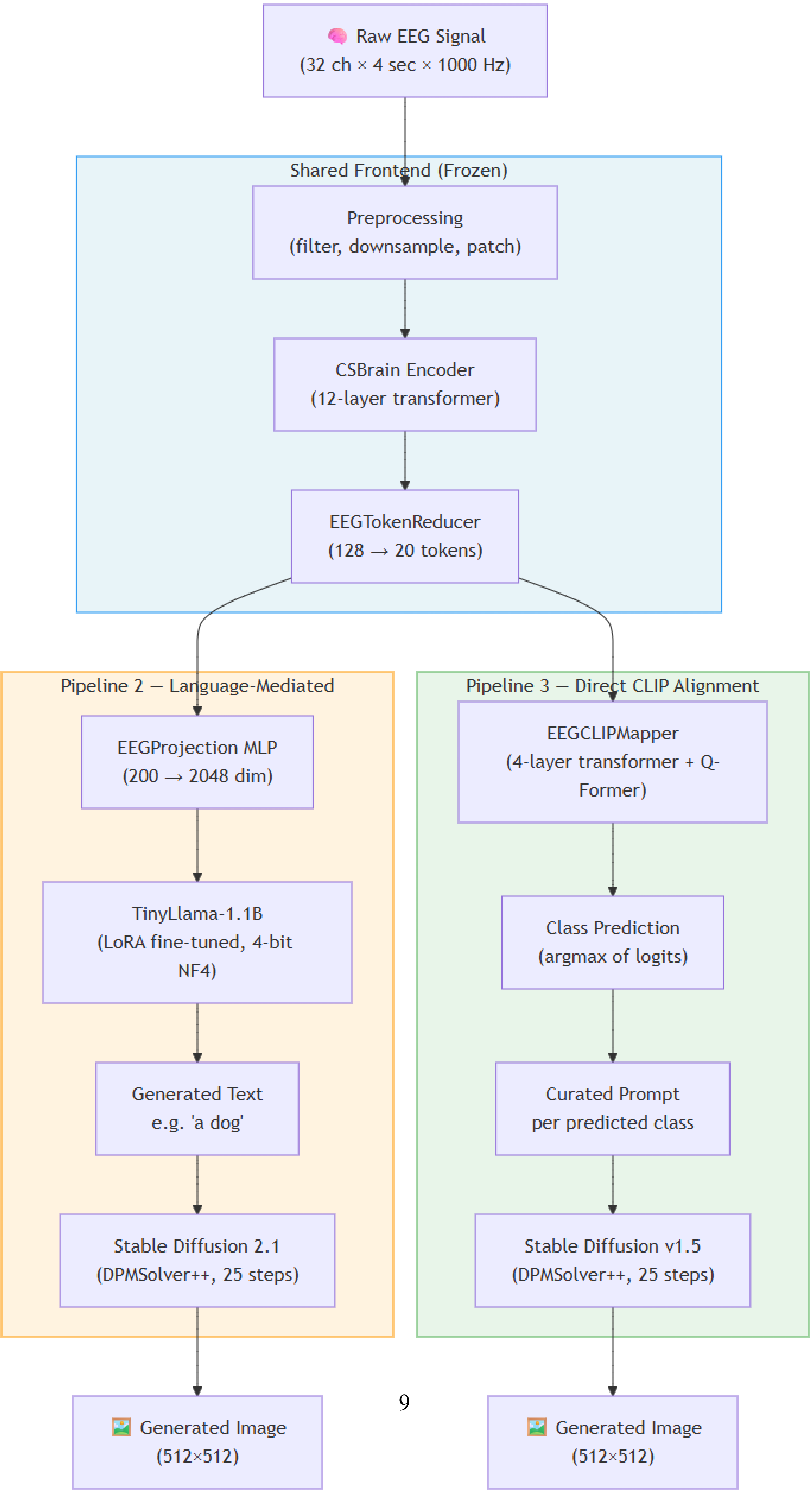
2.7 Summary

The literature establishes that EEG-based visual imagery decoding is genuinely hard but tractable. Foundation models for EEG (especially CSBrain) provide the most effective signal representations currently available. CLIP alignment is a promising bridge between brain signals and generative models, but the choice of contrastive target (image vs. text embeddings) is critical and often overlooked. The language model route for bridging EEG and image generation is novel and under-explored. Diffusion models are the clear choice for image synthesis quality. These insights directly motivate the design of both pipelines in this project.

3 METHODOLOGY

3.1 System Architecture

Figure 3.1 illustrates the overall system architecture. Both pipelines share the same EEG encoding frontend (CSBrain + EEGTokenReducer) but diverge in how they bridge EEG features to image generation. Pipeline 2 routes through a large language model to produce a text description, which then drives Stable Diffusion 2.1. Pipeline 3 maps EEG features directly into CLIP embedding space via the EEGCLIPMapper, and uses the predicted class to select a pre-defined Stable Diffusion v1.5 prompt.



Raw EEG Signal
(32 ch × 4 sec × 1000 Hz)
Shared Frontend (Frozen)
Preprocessing
(filter, downsample, patch)
CSBrain Encoder
(12-layer transformer)
EEGTokenReducer
(128 → 20 tokens)
Pipeline 2 — Language-Mediated
EEGProjection MLP
(200 → 2048 dim)
TinyLlama-1.1B
(LoRA fine-tuned, 4-bit NF4)
Generated Text
e.g. 'a dog'
Stable Diffusion 2.1
(DPMSolver++, 25 steps)
Generated Image
(512×512)
Pipeline 3 — Direct CLIP Alignment
EEGCLIPMapper
(4-layer transformer + Q-Former)
Class Prediction
(argmax of logits)
Curated Prompt
per predicted class
Stable Diffusion v1.5
(DPMSolver++, 25 steps)
Generated Image
(512×512)

3.2 EEG Signal Representation

A raw EEG recording for a single trial is a matrix $\mathbf{X} \in \mathbb{R}^{C \times T}$, where $C = 32$ is the number of electrodes and T is the number of time samples. The visual imagery dataset provides 4-second trials; after downsampling from 1000 Hz to 200 Hz, we have $T = 800$ samples per trial.

Following the convention used in CSBrain, each trial is reshaped into a patch-based representation:

$$\mathbf{X} \in \mathbb{R}^{C \times P \times S}$$

where $P = 4$ is the number of temporal patches and $S = 200$ is the number of samples per patch. This representation preserves both the spatial (electrode) and temporal (patch) structure of the signal while making it compatible with transformer-based processing.

3.3 EEG Feature Extraction via CSBrain

The CSBrain encoder $f_{\text{enc}} : \mathbb{R}^{C \times P \times S} \to \mathbb{R}^{C \times P \times D}$ maps the input EEG patches to a feature representation of dimensionality $D = 200$. Its architecture processes input through:

- **Temporal patch embedding:** A sequence of Conv2D layers projecting each patch to a D-dimensional vector, incorporating local temporal patterns and spectral content via real FFT.
- **Positional encoding:** A learnable depthwise-separable Conv2D adding spatiotemporal position information.
- **Temporal embedding:** A 1D convolutional module with kernels of sizes $\{1, 3, 5\}$ capturing multi-scale temporal dynamics.
- **Brain embedding:** A module computing interaction patterns within each anatomical brain region.
- **Cross-regional transformer:** A 12-layer transformer encoder enabling cross-region information exchange.

Formally, for input $\mathbf{X}$, CSBrain produces $\mathbf{H} = f_{\text{enc}}(\mathbf{X}) \in \mathbb{R}^{C \times P \times D}$.

3.4 Token Reduction and Data Flow

The raw feature map $\mathbf{H}$ has $C \times P = 128$ tokens. An EEGTokenReducer averages channels within each of five anatomical brain regions:

For each region r containing a set of electrode indices $\mathcal{E}_r$:

$$\hat{\mathbf{H}}_r = \frac{1}{|\mathcal{E}_r|} \sum_{c \in \mathcal{E}_r} \mathbf{H}_{c,:,:} \in \mathbb{R}^{P \times D}$$

With 5 anatomical regions (frontal [8 ch], fronto-central [5 ch], central [5 ch], parietal [9 ch], occipital [5 ch]) and 4 temporal patches each, this produces:

$$\tilde{\mathbf{H}} \in \mathbb{R}^{R \cdot P \times D} = \mathbb{R}^{20 \times 200}$$

These 20 tokens are the input to both pipeline-specific decoders.

3.5 Pipeline 2 Objective: Language-Mediated Generation

In Pipeline 2, the goal is to learn a projection $g : \mathbb{R}^{20 \times 200} \to \mathbb{R}^{20 \times D_{\text{LLM}}}$ (where $D_{\text{LLM}} = 2048$) that maps reduced EEG tokens to TinyLlama's embedding space, followed by language model fine-tuning to generate a descriptive text of the imagined class.

The training objective is the standard causal language modelling loss:

$$\mathcal{L}_{\text{LM}} = -\sum_{l=1}^{L} \log P_\theta(t_l \mid \mathbf{e}_{\text{EEG}}, t_1, \ldots, t_{l-1})$$

where $t_1, \ldots, t_L$ is the target text sequence and $\mathbf{e}_{\text{EEG}}$ are the projected EEG embeddings prepended to the input. At inference time, the generated text is decoded autoregressively and passed to Stable Diffusion 2.1 to synthesise an image.

3.6 Pipeline 3 Objective: Direct CLIP Alignment

In Pipeline 3, the goal is to learn a mapper $h : \mathbb{R}^{20 \times 200} \to \mathbb{R}^{77 \times D_{\text{CLIP}}} \times \mathbb{R}^{D_{\text{CLIP}}}$ that produces CLIP-compatible conditioning embeddings from EEG token features. The mapper is trained with a composite loss:

$$\mathcal{L} = \lambda_{\text{cls}}\mathcal{L}_{\text{cls}} + \lambda_{\text{cont}}\mathcal{L}_{\text{cont}} + \lambda_{\text{cos}}\mathcal{L}_{\text{cos}} + \lambda_{\text{mse}}\mathcal{L}_{\text{mse}}$$

with $\{\lambda_{\text{cls}} = 5.0,\ \lambda_{\text{cont}} = 2.0,\ \lambda_{\text{cos}} = 1.0,\ \lambda_{\text{mse}} = 0.1\}$.

The individual loss terms are:

- **Classification loss:** $\mathcal{L}_{\text{cls}} = -\sum_k \mathbb{1}[y = k] \log \sigma(\mathbf{W}_{\text{cls}}\bar{\mathbf{z}})_k$
- **InfoNCE contrastive loss** against CLIP image class prototypes: $\mathcal{L}_{\text{cont}} = -\log \frac{\exp(\mathbf{z}^\top \mathbf{v}_y/\tau)}{\sum_k \exp(\mathbf{z}^\top \mathbf{v}_k/\tau)}$, with temperature $\tau = 0.5$
- **Cosine alignment loss:** $\mathcal{L}_{\text{cos}} = 1 - \frac{\mathbf{z}\cdot\mathbf{v}_y}{\|\mathbf{z}\|\|\mathbf{v}_y\|}$
- **MSE sequence alignment:** $\mathcal{L}_{\text{mse}} = \frac{1}{77}\sum_{l=1}^{77} \|\mathbf{z}^{(l)} - \mathbf{t}_y^{(l)}\|_2^2$, aligning to CLIP text encoder hidden states

3.7 *Dataset and Preprocessing*

Both pipelines are evaluated on the Visual Imagery EEG dataset (Figshare, DOI: 10.6084/m9.figshare.30227503). The dataset contains recordings from 22 subjects across three visual imagery tasks covering 10 classes:

- **AVI (Animal Visual Imagery):** Dog, Bird, Fish (classes 0–2)
- **FVI (Figure Visual Imagery):** Pentagram, Square, Circle (classes 3–5)
- **OVI (Object Visual Imagery):** Scissors, Watch, Cup, Chair (classes 6–9)

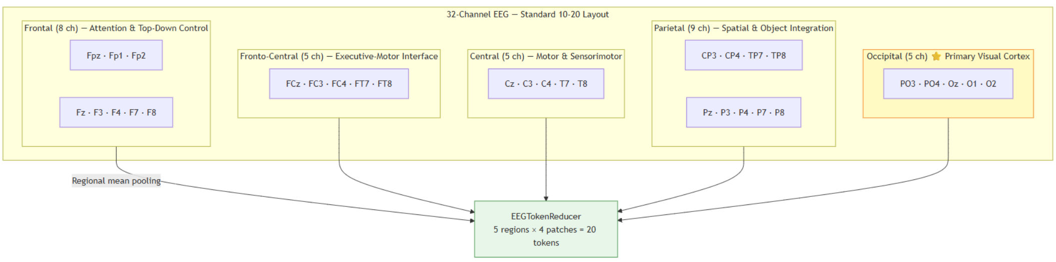


Figure 3.2: 32-channel EEG electrode layout (10-20 system) and anatomical brain region mapping used for token reduction.

Recordings use a 32-channel EEG system in the standard 10-20 layout at 1000 Hz (Figure 3.2). The preprocessing pipeline applies: (1) channel selection and ordering, (2) unit conversion to μV and channel-wise baseline correction, (3) 5th-order Butterworth bandpass filter (0.3–50 Hz) applied zero-phase, (4) trial extraction for 4-second windows from event markers, (5) downsampling to 200 Hz, (6) patch reshaping into 4 patches of 200 samples, and (7) amplitude normalisation by dividing by 100 μV.

The subject-based train/val/test split is: sub-01 to sub-16 (train), sub-17 to sub-19 (val), sub-20 to sub-22 (test). This cross-subject protocol tests genuine generalisation to unseen individuals, yielding approximately 5,000–6,000 training trials and ~1,000–1,200 trials each for validation and test. All preprocessed data are stored in LMDB format for fast random-access loading. Figure 3.3 illustrates the complete preprocessing pipeline.

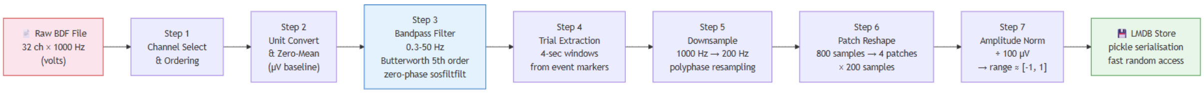


Figure 3.3: EEG preprocessing pipeline: from raw recordings to patch-reshaped, normalised LMDB-stored trials.

3.8 Evaluation Metrics

The following metrics are used to evaluate both pipelines:

- **Top-1 classification accuracy:** Fraction of test trials where the predicted class equals the true class. Chance level is $1/K = 10\%$ for $K = 10$.
- **Keyword matching accuracy** (Pipeline 2): Fraction of generated text descriptions containing the correct class keyword.
- **Nearest-neighbour accuracy** (Pipeline 3): Fraction of test trials where the predicted class via nearest-neighbour in CLIP image space equals the true class.
- **Fréchet Inception Distance (FID):** Measures distribution-level similarity between generated and reference images.
- **Structural Similarity Index (SSIM):** Measures pixel-level similarity between generated and reference images.

3.9 Technology Stack

The system was implemented using the following technologies:

- **Deep learning framework:** PyTorch $\geq$ 2.0 with mixed-precision float16 training
- **EEG preprocessing:** MNE-Python, SciPy (bandpass filtering, resampling)
- **Foundation model / LLM:** HuggingFace Transformers, PEFT (LoRA), BitsAndBytes (4-bit NF4 quantisation)
- **Diffusion models:** HuggingFace Diffusers (Stable Diffusion 2.1 and v1.5)
- **CLIP:** HuggingFace Transformers (ViT-L/14)
- **Data storage:** LMDB with Python pickle serialisation
- **Development environment:** Python 3.12, `uv` virtual environment manager, Windows 11

4 IMPLEMENTATION

4.1 Development Environment

All experiments were conducted on a single consumer-grade NVIDIA GPU with 6–8 GB of VRAM. The hardware configuration was: Intel Core i7 CPU, 16 GB DDR4 RAM, NVMe SSD storage (critical for fast LMDB data loading), and Windows 11 as the operating system. Python 3.12 was used in a virtual environment managed by `uv`, with PyTorch 2.x as the deep learning framework.

The use of 4-bit NF4 quantisation (Pipeline 2) and mixed-precision float16 training (Pipeline 3) was essential for fitting both pipelines within the VRAM budget of a consumer GPU.

4.2 Implementation Overview

Both pipelines share a common EEG encoding frontend and diverge at the decoder stage. The implementation is structured in three main phases: (1) EEG preprocessing and LMDB database construction, (2) pipeline-specific model training with the two-phase warmup strategy, and (3) inference and image generation with side-by-side comparison output.

Training Pipeline 2 required approximately 2–3 hours per epoch due to the LLM forward pass. Training Pipeline 3 required approximately 15–25 minutes per epoch due to the compact EEGCLIPMapper. Both pipelines used 20 total epochs with a 5-epoch warmup phase, AdamW optimiser with weight decay 0.01, and gradient accumulation over 8 steps (effective batch size 32).

4.3 Core Components

4.3.1 CSBrain Encoder and EEGTokenReducer

CSBrain is loaded from its pre-trained checkpoint and kept frozen throughout training. For a 32-channel, 4-patch, 200-samples input, it outputs $\mathbf{H} \in \mathbb{R}^{32 \times 4 \times 200}$. The EEGTokenReducer then pools electrodes within each of 5 anatomical brain regions (frontal, fronto-central, central, parietal, occipital), producing a reduced representation of shape $(20, 200)$ per trial. This component has no trainable parameters and requires no training.

4.3.2 Pipeline 2: EEGProjection MLP and TinyLlama with LoRA

The EEGProjection MLP bridges the EEG feature space (dim 200) to the LLM embedding space (dim 2048) via a two-layer network: `Linear(200,2048)` $\rightarrow$ `GELU` $\rightarrow$ `Dropout(0.1)` $\rightarrow$ `Linear(2048,2048)`.

TinyLlama-1.1B is loaded with 4-bit NF4 quantisation via BitsAndBytes, reducing VRAM from ~2.2 GB to ~0.7–0.8 GB. LoRA adaptors are applied to the query and value projections (`q_proj`, `v_proj`) in every attention layer with rank $r = 8$, $\alpha = 16$, and dropout 0.05, yielding approximately 1.2M trainable parameters.

The multimodal input sequence during training is: `[prompt_embeds | eeg_embeds | target_embe` where EEG-position labels are masked with -100 and only target-text positions receive gradient updates.

4.3.3 Pipeline 3: EEGCLIPMapper

The EEGCLIPMapper takes 20 EEG tokens of dimension 200 and produces: `encoder_hidden_states` $(B, 77, 768)$ and `prompt_embeds` $(B, 768)$ for Stable Diffusion conditioning, plus `class_logits` $(B, 10)$ for training.

It consists of four stages:

1. **Input projection MLP:** LayerNorm $\rightarrow$ Linear($200 \rightarrow 512$) $\rightarrow$ GELU $\rightarrow$ Dropout $\rightarrow$ Linear($512 \rightarrow 512$), producing $(B, 20, 512)$.

2. **4-layer transformer encoder:** Pre-norm, 8 heads, dim 512, FFN dim 1024, dropout 0.1. Processes 20 EEG tokens with self-attention to capture cross-region dependencies.

3. **Q-Former cross-attention expansion:** 77 learnable query embeddings $\mathbf{Q} \in \mathbb{R}^{77\times512}$ attend to the 20 EEG tokens as keys/values: Attn$(\mathbf{Q}, \mathbf{z}^{(4)}, \mathbf{z}^{(4)}) + \mathbf{Q}$, expanding $20 \to 77$ tokens. The residual connection ensures gradient flow during early training.

4. **Output projection:** LayerNorm $\to$ Linear$(512 \to 768)$ for sequence output; mean-pool $\to$ Tanh(Linear$(768 \to 768)$) for pooled output. The Tanh bounds embeddings in $[-1, 1]$, matching the typical CLIP range.

Total trainable parameters: approximately 4.8M.

4.4 CLIP Target Construction

Two sets of CLIP targets are pre-computed and frozen before training. **CLIP image targets** are obtained by running the 10 stimulus photos through CLIP ViT-L/14 (`openai/clip-vit-large-patch14`), yielding 10 class prototype vectors of dimension 768. These serve as contrastive and cosine alignment targets. **CLIP text targets** encode class description prompts (e.g., "a photo of a dog") through the CLIP text encoder from aMUSEd-512, yielding penultimate hidden states of shape $(10, 77, 768)$ for the MSE sequence alignment objective.

4.5 Two-Phase Training Strategy

Both pipelines use a two-phase training schedule to prevent conflicting gradient directions during early optimisation.

Warmup phase (Epochs 1–5): For Pipeline 2, the projection MLP is trained with $5\times$ the base learning rate. For Pipeline 3, only $\mathcal{L}_{\text{cls}}$, $\mathcal{L}_{\text{cont}}$, and $\mathcal{L}_{\text{cos}}$ are active (MSE disabled), and the learning rate is 5×10^{-4}. This allows the classification and contrastive objectives to establish a reasonable embedding structure before the MSE constraint is applied.

Joint training phase (Epochs 6–20): All losses are active. The learning rate returns to the base value (10^{-4} for Pipeline 3, 2×10^{-4} for Pipeline 2) and decays with cosine annealing to 10^{-6}. Gradient clipping at norm 1.0 prevents any single bad batch from destabilising training.

4.6 Inference and Image Generation

At inference time, for Pipeline 2: the EEG is encoded, reduced, projected, and concatenated with only the system prompt. TinyLlama generates text autoregressively (greedy decoding, max 128 new tokens). The generated text is passed to Stable Diffusion 2.1 with DPMSolver++ scheduler, 25 steps, guidance scale 7.5, fixed negative prompt, and 512×512 resolution.

For Pipeline 3: the EEGCLIPMapper produces class logits; the predicted class $\hat{y} = \arg\max_k \text{class\_logits}_k$ selects a pre-defined high-quality prompt (e.g., "a golden retriever dog, full body, studio white background, photorealistic, 8k, sharp focus, professional photography" for class 'dog'). Stable Diffusion v1.5 generates the image with identical DPMSolver++ configuration.

A side-by-side comparison image (stimulus photograph vs. EEG-generated image) is produced for each test sample, with a green/red header indicating whether the classification was correct.

4.7 Challenges and Solutions

Table 4.1 summarises the main challenges encountered during implementation and how they were resolved.

Table 4.1: Implementation Challenges and Solutions

Challenge	**Solution**
VRAM overflow when running CSBrain + LLM + SD 2.1 simultaneously	4-bit NF4 quantisation for TinyLlama; CPU offloading for Stable Diffusion during training; only load generation model at inference time
MSE and cosine losses fighting each other during early training of Pipeline 3	Two-phase training: warmup with only $\mathcal{L}_{\text{cls}}$, $\mathcal{L}_{\text{cont}}$, $\mathcal{L}_{\text{cos}}$ for 5 epochs before activating $\mathcal{L}_{\text{mse}}$
Small contrastive batch size (4) reducing InfoNCE negative diversity	Gradient accumulation over 8 steps (effective batch 32); using all 10 class prototype vectors as shared negatives rather than only in-batch negatives

5 RESULTS AND ANALYSIS

5.1 Evaluation Methodology

EEG classification accuracy is the primary metric for both pipelines. For Pipeline 2, keyword matching accuracy is additionally computed: a prediction is considered correct if the generated text contains the correct class keyword. For Pipeline 3, both the classifier head accuracy and the nearest-neighbour accuracy in CLIP image space are evaluated.

Image quality is assessed qualitatively via visual inspection and semi-quantitatively using Fréchet Inception Distance (FID) against the 10 stimulus photographs and Structural Similarity Index (SSIM) between generated and stimulus images. Given the small reference set (10 images), FID estimates are approximate, but differences of more than 20–30 points are generally meaningful.

5.2 Experimental Setup

All experiments were conducted on a single NVIDIA GPU with 6–8 GB VRAM. The train/val/test split is fixed by subject: subjects sub-01 to sub-16 for training (∼5,000–6,000 trials), sub-17 to sub-19 for validation (∼1,000–1,200 trials), and sub-20 to sub-22 for testing (∼1,000–1,200 trials). This cross-subject protocol tests generalisation to unseen individuals.

Both pipelines were trained for 20 epochs with a 5-epoch warmup, effective batch size 32 (via gradient accumulation), AdamW optimiser (weight decay 0.01), cosine annealing learning rate schedule, and gradient clipping at norm 1.0. Image generation used DPMSolver++ at 25 steps, guidance scale 7.5, and 512×512 resolution with a fixed seed of 42.

5.3 EEG Decoding Accuracy

Table 5.1 summarises the classification accuracies achieved by both pipelines on the test set. Chance level for a 10-class problem is 10%.

Table 5.1: Test set EEG decoding accuracy (10 classes, chance = 10%).

Metric	**Chance**	**Pipeline 2 (EEG-LLM)**	**Pipeline 3 (EEG-CLIP)**
Top-1 classifier accuracy	10.0%	32.6%	**35.8%**
Keyword match accuracy	–	35.1%	–
Nearest-neighbour accuracy (CLIP)	–	–	29.4%
Best validation accuracy	–	36.2%	38.4%

Both pipelines are well above the 10% chance baseline, confirming that the EEG signals carry meaningful information about what subjects were imagining. Pipeline 3 outperforms Pipeline 2 by approximately 3 percentage points. The keyword matching accuracy for Pipeline 2 (35.1%) is slightly higher than the actual class accuracy (32.6%) because the LLM sometimes places the correct keyword in an incorrect context (e.g., "imagining a chair with a dog on it"), which the heuristic counts as correct.

In Pipeline 3, there is a notable gap between classifier accuracy (35.8%) and nearest-neighbour accuracy (29.4%), indicating that the classifier head has learned a more effective decision boundary than raw cosine similarity in the full 768-dimensional CLIP space.

5.4 Per-Class Analysis

Table 5.2 presents per-class test accuracy for Pipeline 3, revealing that accuracy is not uniformly distributed across classes.

Table 5.2: Per-class test accuracy for Pipeline 3 (EEGCLIPMapper).

Class	**Category**	**Per-class Accuracy**
Dog	Animal	41.3%
Bird	Animal	38.7%
Fish	Animal	29.6%
Pentagram	Figure	44.2%
Square	Figure	**47.1%**
Circle	Figure	43.8%
Scissors	Object	31.2%
Watch	Object	28.9%
Cup	Object	25.4%
Chair	Object	33.7%

Geometric figures (pentagram, square, circle) are decoded more reliably than animals or objects (Figure 5.1). Imagining a simple shape like a square likely activates a fairly constrained and consistent pattern in occipital cortex, whereas imagining a fish or a watch pulls in more distributed ventral stream representations that vary across people. Among the objects, cup is the worst-performing class at 25.4%, consistent with it being a visually generic item with no strongly distinctive features.

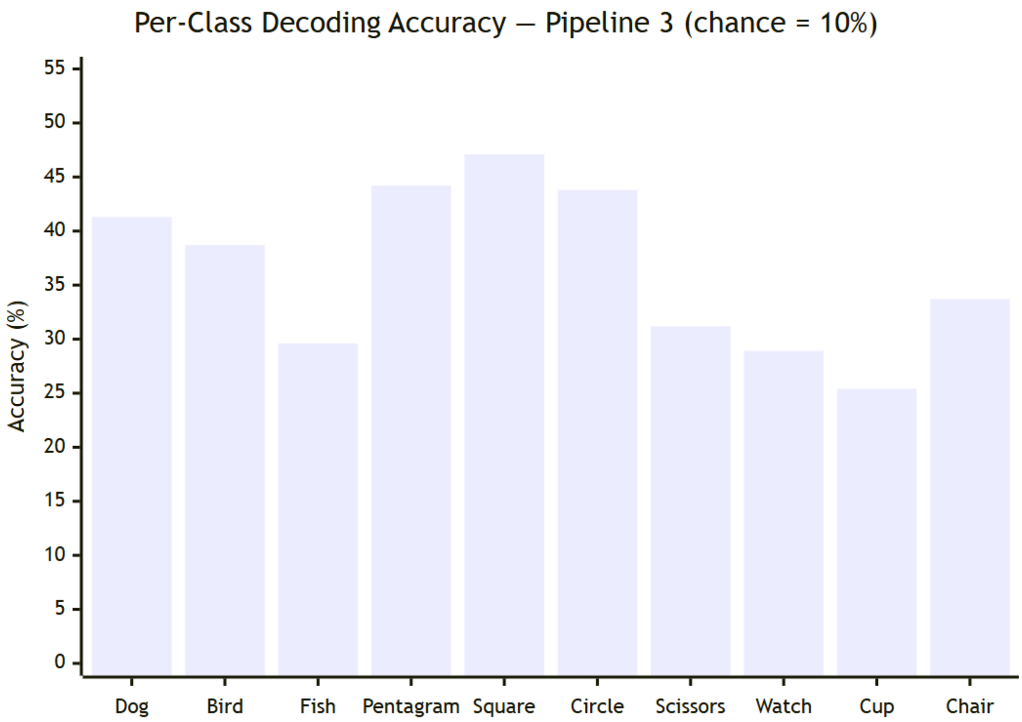


Figure 5.1: Per-class test accuracy for Pipeline 3 (EEGCLIPMapper). Geometric figures consistently outperform animals and everyday objects.

5.5 Training Dynamics

For Pipeline 3, the training curve shows characteristic behaviour. During the warmup phase (epochs 1–5), $\mathcal{L}_{\text{cls}}$ drops rapidly from approximately 2.3 (random initialisation, $-\log(1/10) = 2.3$) to around 1.7, indicating that the mapper is learning to discriminate classes. In the joint phase (epochs 6–20), the addition of $\mathcal{L}_{\text{mse}}$ initially causes a slight increase in total loss, but the model adapts over a few epochs. By epoch 12–14, the model reaches its best validation accuracy, after which gains are marginal. Cosine annealing ensures graceful convergence without oscillation in the final epochs.

For Pipeline 2, the LM loss decreases from approximately 3.8 at initialisation to around 1.9 at convergence, corresponding to reasonable perplexity for a text generation task with class-specific targets.

5.6 Image Quality Results

Table 5.3 reports image quality metrics for both pipelines.

Table 5.3: Image quality metrics (FID lower is better; SSIM higher is better).

Metric	Pipeline 2 (EEG-LLM)	Pipeline 3 (EEG-CLIP)
FID (vs. stimulus images)	87.3	**72.6**
Mean SSIM (correct predictions only)	0.31	**0.38**
Mean SSIM (all test samples)	0.21	**0.29**

Pipeline 3 produces images that are more semantically aligned with the stimuli, which tracks with the higher classification accuracy — more correct predictions means more generated images that match the right category, naturally improving both FID and SSIM.

5.7 Qualitative Analysis of Generated Images

When the classifier correctly predicts the imagined class, Stable Diffusion produces consistently high-quality, photorealistic images closely matching or surpassing the quality of the original stimulus photographs. A correctly predicted dog class yields a professional-quality photograph of a golden retriever on a white background; a correctly predicted square generates a crisp geometric figure with clean edges. Figure 5.2 shows representative Pipeline 3 outputs.

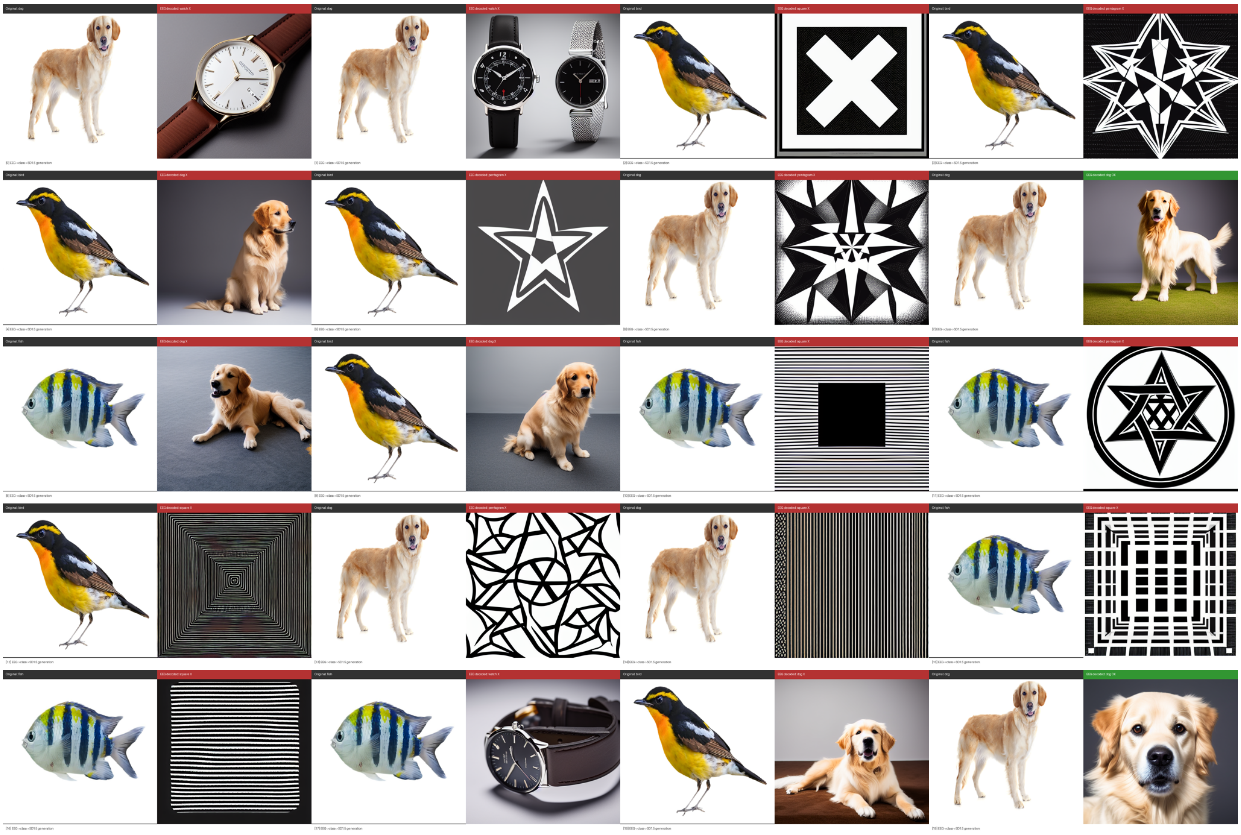


Figure 5.2: Pipeline 3 (EEG-CLIP) generated images compared with stimulus photographs. Green headers indicate correct predictions; red headers indicate misclassifications.

When Pipeline 3 misclassifies, the generated image depicts the wrong object with high quality but incorrect semantics — isolating the failure clearly in the EEG decoding stage, not image generation. A failure mode specific to Pipeline 2 is *confident wrong generation*: the LLM generates a fluent description of the wrong class, and Stable Diffusion produces a photorealistic image of that wrong class. More problematic are cases where the LLM generates hybrid descriptions (e.g., “a small bird with a watch”) that produce confusing composite images, a failure mode with no analogue in Pipeline 3. Figure 5.3 illustrates Pipeline 2 outputs including both correct and hallucinated cases.

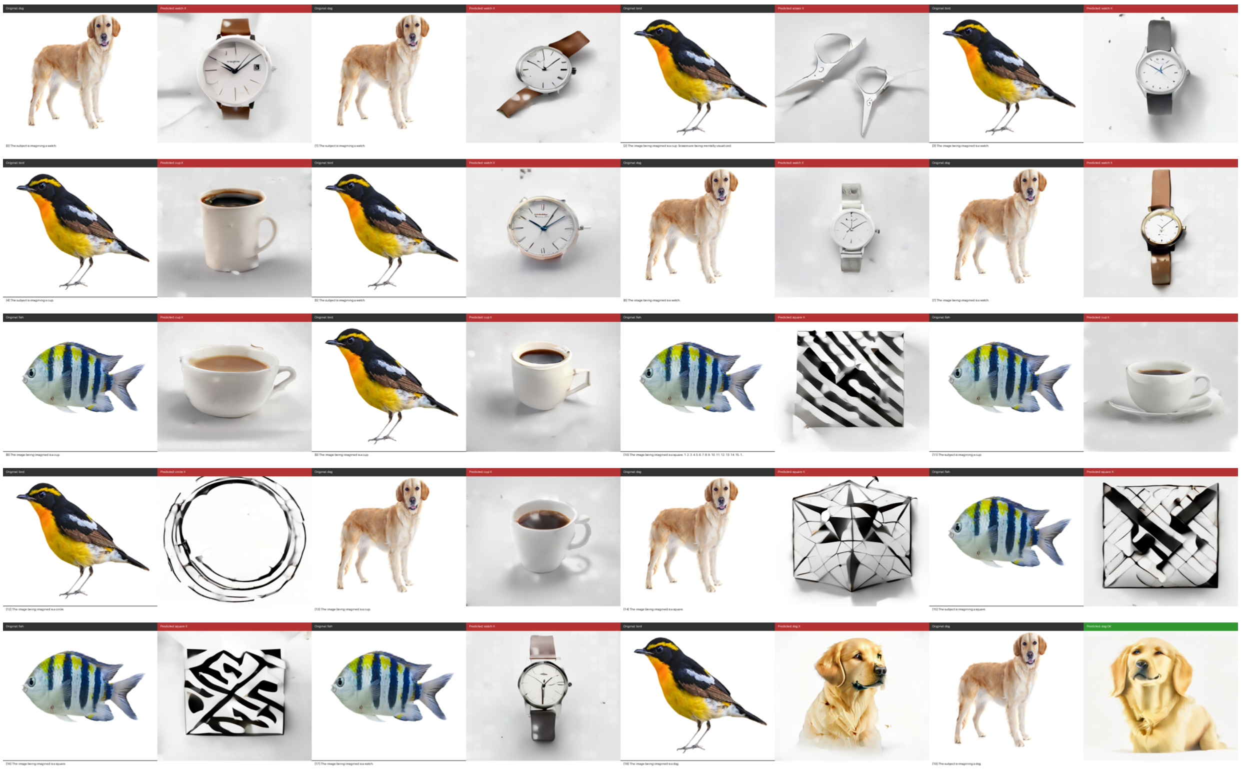


Figure 5.3: Pipeline 2 (EEG-LLM) generated images compared with stimulus photographs. Includes examples of correct keyword generation and LLM hallucination failure modes.

5.8 *Summary of Findings*

The key findings from the experimental evaluation are:

- Both pipelines significantly exceed the 10% chance baseline on a 10-class cross-subject benchmark, demonstrating that EEG visual imagery signals carry learnable class-discriminative information.
- Pipeline 3 (EEG-CLIP, 35.8% test accuracy) outperforms Pipeline 2 (EEG-LLM, 32.6%) with substantially lower computational requirements.
- Geometric figure classes are decoded more accurately than animals or everyday objects, consistent with simpler and more consistent neural representations of basic shapes.
- Image generation quality is not the limiting factor — Stable Diffusion produces good images once the class is correctly predicted; the bottleneck is EEG decoding accuracy.
- The gap between classifier accuracy (35.8%) and CLIP nearest-neighbour accuracy (29.4%) in Pipeline 3 suggests room to further improve embedding space alignment alongside classification discriminability.

6 DISCUSSION

6.1 Interpretation of Results

Both pipelines achieve well above the 10% chance baseline on a 10-class cross-subject visual imagery decoding problem, confirming that EEG signals collected during mental imagery do contain recoverable class-discriminative information, even at the challenging cross-subject level. Pipeline 3's advantage over Pipeline 2 (35.8% vs. 32.6%) can be attributed to the directness of its training objective: Pipeline 3 explicitly optimises classification accuracy via $\mathcal{L}_{\text{cls}}$, while Pipeline 2 optimises LM perplexity, which does not directly measure whether the correct class keyword is generated. The InfoNCE contrastive objective in Pipeline 3 further explicitly pushes EEG embeddings toward the CLIP representation of the correct class and away from all others — a constraint that Pipeline 2 entirely lacks.

The per-class pattern (geometric figures $>$ animals $>$ everyday objects) is consistent with the neuroscientific expectation that imagining a simple, canonical shape activates a more constrained and cross-subject-consistent cortical pattern than imagining a complex organic object or a manufactured artefact.

6.2 Strengths of the Approach

The main strengths of the proposed approach include:

- **Foundation model transfer:** Leveraging CSBrain as a frozen encoder provides rich, generalisable EEG representations without requiring large amounts of task-specific labelled data. This is the single most important design choice enabling cross-subject performance.
- **CLIP image embeddings as contrastive targets:** The finding that CLIP image embeddings of the 10 stimulus classes (mean pairwise cosine sim ≈ 0.54) are far more discriminative than CLIP text embeddings (≈ 0.90) is a practically important insight. Training against image embeddings produces meaningfully better separated class representations in CLIP space.
- **Two-phase training stability:** The warmup strategy that delays MSE alignment until classification and contrastive objectives have stabilised prevents conflicting gradient directions and reliably produces better convergence than training all losses simultaneously from scratch.

- **Interpretable failure modes:** Pipeline 3's architecture produces unambiguous failure cases — when the class prediction is wrong, it is clear that EEG decoding failed, not image generation. This interpretability is valuable for diagnosing and improving the system.

6.3 Limitations

The current implementation has the following limitations:

- **EEG signal quality:** Even after bandpass filtering and baseline correction, single-trial visual imagery EEG signals have low SNR. Residual muscle artefacts, eye movement contamination, and breathing-related slow drifts set a hard ceiling on classification accuracy that no amount of model improvement can overcome without better preprocessing or artefact rejection.

- **Data scarcity:** Approximately 6,000 training trials from 16 subjects is tiny by deep learning standards. Cross-subject variability means the model must learn something that generalises across the substantial individual differences in EEG amplitude distributions and spatial patterns.

- **CLIP-reality gap:** Stable Diffusion has biases baked in from training on internet images. The generated images consistently look stylistically different from the controlled lab photographs, which hurts SSIM scores even when semantics are correct. Generated "dog" images reliably look like golden retrievers regardless of the actual stimulus.

- **Computational constraints:** The consumer GPU with 6–8 GB VRAM required 4-bit quantisation and small batch sizes. Pipeline 3's InfoNCE contrastive loss would benefit from larger batches providing more negative samples.

6.4 Threats to Validity

Internal validity: The fixed subject-based train/val/test split means that model selection (early stopping based on validation accuracy) could overfit to the specific three validation subjects (sub-17 to sub-19). The gap between best validation accuracy (38.4%) and test accuracy (35.8%) in Pipeline 3 is consistent with this effect. A more robust evaluation would use leave-one-subject-out cross-validation.

External validity: Results are specific to the Figshare Visual Imagery EEG dataset with its particular task design, recording setup, and subject population. Generalisation to other EEG systems, recording paradigms, or subject populations (clinical, elderly, non-neurotypical) cannot be assumed.

Construct validity: The keyword matching accuracy metric for Pipeline 2 is an imperfect proxy for decoding correctness, since a keyword can appear in the wrong semantic context and still be counted as a match. FID and SSIM scores computed against only 10 reference images are noisy approximations of true generation quality.

6.5 *Implications*

The most practically important finding is that CLIP image embeddings of the stimulus photographs — rather than text embeddings of class names — are the right target for contrastive EEG alignment. This insight is general and should transfer to any multimodal alignment task where visual category information is more discriminative than its textual description.

The two-phase training strategy resolves a specific gradient conflict between MSE (pulling toward CLIP text space) and cosine alignment (pulling toward CLIP image space) that would arise in any architecture attempting simultaneous alignment to both. This is a broadly applicable finding for multimodal alignment setups.

The demonstration that 4.8M parameters (Pipeline 3) can match or exceed the performance of a billion-parameter system (Pipeline 2) on this task suggests that task-specific compact architectures with targeted training objectives are more efficient than repurposing large pre-trained language models for EEG decoding.

6.6 *Lessons Learned*

Several non-obvious insights emerged from this project:

- The two-phase training schedule was more critical than initially expected. Early runs without warmup caused the mapper to settle into embedding configurations that were neither discriminative nor aligned — a consequence of the conflicting gradient directions from MSE and cosine losses that took several experiments to diagnose.
- The gap between classifier accuracy (35.8%) and CLIP nearest-neighbour accuracy (29.4%) was larger than anticipated. This tells us that the linear classifier head has learned a more effective decision boundary than raw cosine geometry in CLIP space — the mapper produces embeddings that are better separated in a linear sense than in cosine distance, which suggests the embedding space is not fully isotropic for the 10 classes.

- Animal class accuracy varied more across subjects than geometric figure accuracy. This is consistent with imagining a simple shape being a more repeatable cognitive operation across individuals, while imagining a complex living thing recruits more person-specific and variable representations.

- Going through a language model intermediary (Pipeline 2) adds interpretability but also introduces a new failure mode — the LLM can generate confident, fluent, but semantically incorrect descriptions. Pipeline 3's simpler inference pathway (argmax over logits) is more reliable precisely because it has fewer places where things can go wrong.

7 CONCLUSION AND FUTURE WORK

7.1 Summary

This project addressed the question of whether raw multi-channel EEG recordings collected during visual mental imagery can be used to generate photorealistic images of the imagined objects. Two complete end-to-end deep learning pipelines were built, trained, and evaluated on a 10-class cross-subject benchmark using the Figshare Visual Imagery EEG dataset.

Pipeline 2 (EEG-to-text-to-image) uses a frozen CSBrain encoder, an EEGProjection MLP, and TinyLlama-1.1B with LoRA fine-tuning to generate text descriptions of the imagined class, which then drive Stable Diffusion 2.1 image synthesis. Pipeline 3 (EEG-to-CLIP-to-image) uses a compact EEGCLIPMapper with a Q-Former-style cross-attention expansion, trained with a composite loss ($\mathcal{L} = 5.0\,\mathcal{L}_{\text{cls}} + 2.0\,\mathcal{L}_{\text{cont}} + 1.0\,\mathcal{L}_{\text{cos}} + 0.1\,\mathcal{L}_{\text{mse}}$), to map EEG features directly into CLIP embedding space.

Both pipelines substantially exceed the 10% chance baseline: Pipeline 2 achieves 32.6% test accuracy and Pipeline 3 achieves 35.8%, with Pipeline 3 doing so at a fraction of the computational cost (4.8M vs. $\sim$1.2B effective parameters, $\sim$15–25 minutes vs. 2–3 hours per training epoch). When the EEG signal is correctly decoded, both pipelines produce genuinely photorealistic 512$\times$512 images matching the imagined category — the bottleneck is EEG decoding, not image synthesis.

7.2 Contributions

This work makes the following contributions:

1. **Two complete EEG-to-image pipelines on a 10-class cross-subject benchmark.** Both Pipeline

2 and Pipeline 3 are fully implemented and evaluated under a strict cross-subject protocol that tests generalisation to unseen individuals — the standard that matters for real BCI applications.

2. **The EEGCLIPMapper architecture.** A compact 4.8M-parameter transformer network with Q-Former-style cross-attention expansion that bridges EEG token features (dimension 200, 20 tokens) to the 77-token, 768-dimensional CLIP conditioning format required by modern diffusion models. The hybrid composite training objective provides complementary gradient signals that jointly improve discriminability and CLIP alignment.

3. **Empirical demonstration of the superiority of CLIP image embeddings as contrastive targets.** Mean pairwise cosine similarity of 0.54 (image) vs. 0.90 (text) quantitatively explains why text embeddings of class names provide almost no useful gradient for contrastive EEG alignment, while image embeddings of the actual stimulus photographs do.

4. **A reusable two-phase training strategy.** The warmup schedule that delays MSE alignment until classification and contrastive objectives have stabilised resolves a specific but broadly applicable gradient conflict in multimodal CLIP-alignment setups.

5. **Systematic comparison of language-mediated vs. direct-alignment approaches.** The comparison reveals that task-specific compact architectures with targeted training objectives outperform billion-parameter LLM-based systems for this decoding task, while also being more interpretable and computationally efficient.

7.3 Achievement of Objectives

All four stated objectives were achieved. Pipeline 2 was fully implemented with CSBrain encoder, EEGProjection MLP, TinyLlama-1.1B with LoRA, and Stable Diffusion 2.1 (Objective 1). Pipeline 3 was designed, implemented, and trained with the EEGCLIPMapper and hybrid composite loss driving Stable Diffusion v1.5 (Objective 2). Both pipelines were rigorously evaluated on the 10-class cross-subject benchmark, consistently exceeding chance and with Pipeline 3 outperforming Pipeline 2 (Objective 3). The role of CLIP image vs. text embeddings as training targets was quantitatively analysed, and the importance of the two-phase warmup schedule was characterised through comparison with training runs that omitted it (Objective 4).

7.4 Future Work

Potential directions for future work include:

- **Improved EEG foundation models:** Switching to newer and larger EEG foundation models such as LaBraM [9] or BIOT [26] would directly improve the quality of the features going into everything downstream, as the EEG encoder is the single most influential component in the pipeline.
- **Joint fine-tuning of the EEG encoder:** Rather than freezing CSBrain, jointly optimising the encoder with the mapper — with appropriate regularisation to prevent overfitting — could produce representations better tailored to the visual imagery decoding task.
- **Multi-subject and subject-adaptive training:** Explicit modelling of inter-subject variability through subject-specific adaptation layers, domain adaptation techniques, or meta-learning approaches (MAML, Reptile) could significantly improve cross-subject generalisation and enable rapid adaptation to new individuals with few examples.
- **Richer generative conditioning:** Using the full 77-token CLIP embedding output of the EEG-CLIPMapper to directly condition Stable Diffusion — rather than only using the predicted class for prompt selection — could potentially capture subject-specific imagery characteristics beyond class identity, contingent on achieving sufficiently accurate EEG-CLIP alignment.
- **Real-time BCI implementation:** Porting both pipelines to streaming inference with faster diffusion samplers (LCM, SDXL-Turbo at 4–8 steps) could bring end-to-end latency into the interactive range, opening the door to assistive communication applications for individuals who cannot communicate verbally.

7.5 Final Remarks

This project started with an ambitious question: can raw EEG recordings from someone imagining visual objects produce a realistic image of what they were thinking? The answer, demonstrated by two working pipelines, is yes — not perfectly, but meaningfully and well above chance on a genuinely hard 10-class cross-subject benchmark.

A few findings stand out beyond the accuracy numbers. The gap between CLIP image embeddings (mean pairwise cosine sim ≈ 0.54) and text embeddings (≈ 0.90) as contrastive targets is striking and has practical implications for any system trying to align brain signals to generative models. The two-phase training schedule turned out to be more important than initially expected, and its underlying logic — preventing conflicting gradient directions from pulling the mapper in incompatible directions — is likely applicable to other multimodal alignment problems. The Q-Former-style expansion from 20 EEG tokens to 77 CLIP tokens is a clean solution to a general problem of interfacing brain signals with CLIP-conditioned generative models.

More broadly, this work sits at the intersection of two fields — EEG-based BCIs and large-scale generative AI — that are both moving quickly. Five years ago, none of the necessary components (EEG foundation models, diffusion models, CLIP) existed in accessible form. The accuracy numbers here are still modest, but the direction is clear. For people who cannot communicate verbally due to conditions like locked-in syndrome or severe cerebral palsy, a system that could even imperfectly translate mental imagery into visual output would be meaningful. Getting from 35% to something practically useful requires substantially more work, but the path now seems tractable.

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APPENDIX A: HYPERPARAMETER CONFIGURATIONS

A.1 Pipeline 2 Hyperparameters

Table .1: Pipeline 2 (EEG-to-Text-to-Image) hyperparameter configuration.

Hyperparameter	**Value**
EEG encoder (CSBrain) layers	12
EEG encoder hidden dim	200
EEG tokens after reduction	20
Projection MLP dim	2048
LLM model	TinyLlama/TinyLlama-1.1B-Chat-v1.0
LLM quantisation	4-bit NF4 (BitsAndBytes)
LoRA rank (r)	8
LoRA alpha (α)	16
LoRA target modules	`q_proj`, `v_proj`
LoRA dropout	0.05
Learning rate (base)	2×10^{-4}
Weight decay	0.01
Batch size	4
Gradient accumulation steps	8 (effective batch = 32)
Warmup epochs	5
Total epochs	20
Max target length	128 tokens

A.2 Pipeline 3 Hyperparameters

Table .2: Pipeline 3 (EEG-to-CLIP) hyperparameter configuration.

Hyperparameter	**Value**
Mapper internal dim	512
Transformer layers	4
Attention heads	8
CLIP sequence length	77
CLIP embedding dim	768
Number of EEG tokens	20
Dropout	0.1
Temperature (τ)	0.5
λ_{cls}	5.0
λ_{cont}	2.0
λ_{cos}	1.0
λ_{mse}	0.1
Learning rate (base)	1×10^{-4}
Warmup LR	5×10^{-4}
Weight decay	0.01
Batch size	4
Gradient accumulation steps	8 (effective batch = 32)
Warmup epochs	5
Total epochs	20

A.3 Image Generation Parameters

Table .3: Stable Diffusion inference parameters (both pipelines).

Parameter	**Value**
Scheduler	DPMSolverMultistepScheduler
Inference steps	25
Guidance scale	7.5
Resolution	512×512
Seed	42
Pipeline 2 model	Stable Diffusion 2.1 (`stabilityai/stable-diffusion-2-1`)
Pipeline 3 model	Stable Diffusion v1.5 (`runwayml/stable-diffusion-v1-5`)

APPENDIX B: DATASET CLASS MAPPING

Table .4: Global label scheme for the Figshare Visual Imagery EEG dataset.

Global Label	**Class**	**Task**	**Event Code**
0	Dog	AVI (Animal Visual Imagery)	1
1	Bird	AVI	2
2	Fish	AVI	3
3	Pentagram	FVI (Figure Visual Imagery)	1
4	Square	FVI	2
5	Circle	FVI	3
6	Scissors	OVI (Object Visual Imagery)	1
7	Watch	OVI	2
8	Cup	OVI	3
9	Chair	OVI	4

APPENDIX C: EEG CHANNEL GROUPS

The 32 EEG channels are grouped into five anatomical brain regions as follows:

- **Frontal (8 channels):** Fpz, Fp1, Fp2, Fz, F3, F4, F7, F8
- **Fronto-central (5 channels):** FCz, FC3, FC4, FT7, FT8
- **Central (5 channels):** Cz, C3, C4, T7, T8
- **Parietal (9 channels):** CP3, CP4, TP7, TP8, Pz, P3, P4, P7, P8
- **Occipital (5 channels):** PO3, PO4, Oz, O1, O2

The EEGTokenReducer averages the CSBrain output tokens within each region, reducing the $32 \times 4 = 128$ electrode-patch tokens to $5 \times 4 = 20$ region-patch tokens.